B. Rising Sand

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: constructive algorithms, greedy, implementation |
https://codeforces.com/problemset/problem/1698/B

There are n piles of sand where the i-th pile has a_i blocks of sand. The i-th pile is called *too tall* if 1 lt i lt n and a_i gt $a_{i-1} + a_{i+1}$. That is, a pile is too tall if it has more sand than its two neighbours combined. (Note that piles on the ends of the array cannot be too tall.)

You are given an integer k. An operation consists of picking k consecutive piles of sand and adding one unit of sand to them all. Formally, pick $1 \le l,r \le n$ such that $r-l+1=k$. Then for all $l \le i \le r$, update a_i gets a_i+1.

What is the **maximum** number of piles that can simultaneously be too tall after some (possibly zero) operations?

Input

The input consists of multiple test cases. The first line contains an integer t ($1 \le t \le 1000$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains two integers n and k ($3 \le n \le 2 \cdot 10^5$; $1 \le k \le n$) — the number of piles of sand and the size of the operation, respectively.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the sizes of the piles.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the maximum number of piles that are simultaneously too tall after some (possibly zero) operations.

Example

Input

```
3
5 2
2 9 2 4 1
4 4
1 3 2 1
3 1
1 3 1
```

Output

```
2
0
1
```

Note

In the first test case, we can perform the following three operations:

- Add one unit of sand to piles 1 and 2: [color*red*3, color*red*10, 2, 4, 1].
- Add one unit of sand to piles 4 and 5: [3, 10, 2, color*red*5, color*red*2].
- Add one unit of sand to piles 3 and 4: [3, 10, color*red*3, color*red*6, 2].

Now piles 2 and 4 are too tall, so in this case the answer is 2. It can be shown that it is impossible to make more than 2 piles too tall.

In the second test case, any operation will increase all piles by 1 unit, so the number of too tall piles will always be 0.

In the third test case, we can increase any pile by 1 unit of sand. It can be shown that the maximum number of too tall piles is 1.